
Additional Problems for Ch 7 HRW – Kinetic energy and Work
 Addisionele Probleme vir Hfst 7 HRW – Kragte 2 Kinetiese energie en Arbeid

	9 th Edition	8 th Edition
Questions	1, 4, 9, 10	3, 2, 9
Problems	1, 5, 8, 19, 29, 31, 49, 50, 57, 61, 65, 70	3, 5, 10, 23, 31, 29, 47, 48, 65, 55, 57, 58

1	3	(a) $v = 2.9 \times 10^7$ m/s. (b) $\Delta K = 6.9 \times 10^{-13}$ J – 4.8×10^{-13} J = 2.1×10^{-13} J.	
5	5	(a) $v_i = 2.4$ m/s. (b) $v_{\text{son}} = 2v_i = 4.8$ m/s.	
8	10	$W = 6.6 \times 10^3$ J.	$W = 5.0 \times 10^3$ J.
19	23	The kinetic energy would have been 25 J greater	
29	31	(a) $W_s = 0.90$ J. (b) $W_s = 2.1$ J. (c) $W_s = 0$ J.	
31	29	(a) $v_f = 6.6$ m/s. (b) $x_f = 4.7$ m.	
49	47	$P = 7.4 \times 10^2$ W.	
50	48	(a) $P = 28$ W (b) $v = 7.5$ m/s.	(a) $P = 28$ W (b) $v = 6$ m/s.
57	65	(a) $F = (2391 \text{ N}) \sin 19.5^\circ = 797$ N. (b) Since there is no change in kinetic energy, the net work on it is zero. (c) $W_g = -1547$ J = -1.55 kJ. (d) The tension vector is everywhere perpendicular to the direction of motion, so its work is zero (since $\cos 90^\circ = 0$). (e) $W_F = -W_g = 1.55$ kJ. (f) Since $\vec{F}$ does not have constant magnitude, we cannot expect $W = \vec{F} \cdot \vec{d}$ to apply.	
61	55	$W = 12 \text{ J} - 18 \text{ J} = -6 \text{ J}$.	
65	57	(a) $ \vec{F} = 98$ N. (c) $W = 3.9$ J.	(b) $d = 0.040$ m. (d) $W = -3.9$ J.
70	58	(a) If $W = 0$, then $c = 6.0$ N. (c) If $W = -18$ J, then $c = 15$ N.	
		(b) If $W = 17$ J, then $c = -2.5$ N.	